

Problem 3 - Linear Regression via Gradient Descent

$$a) f_{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

$$\text{where } \hat{y} = b + w_1 x_1 + \dots + w_m x_m = b + \sum_{i=1}^m w_i x_i$$

$$\Rightarrow \frac{\delta f_{MSE}}{\delta w_j} = \frac{2x_j}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$$

$$b) \frac{\delta f_{MSE}}{\delta w} = \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i$$

$$= \frac{2}{n} x^T (\hat{y} - y)$$
$$= \frac{2}{n} (\hat{y} - y)^T x$$

$$c) \text{ Learning Rate} = 2 \times 10^{-4}$$

$$\text{Training Loss } D_{tr} = 297.1$$

$$\text{Test Loss } D_{te} = 312.3$$

d) Change the gradient descent function to also track loss on the test set. Stop gradient descent if the test loss stops decreasing.